CENTERGUARD · HACKUTD · TECHNICAL BRIEF

How CenterGuard's **predictions work**, and why they are science-based

The physics, published data and statistics behind every number the operator sees, what was validated, and where the model is still an assumption.

Repository Lookahead (pitch name CenterGuard)
Engine `src/domain/engine/` · validation run 2026-11-08
Readers hackathon judges and the team

WHAT THIS DOCUMENT SHOWS

2,048 trials

per response option in round 0 (256 parameter draws × 8 aleatory draws), each a full physics-and-cascade run.

0 numbers

produced by Nemotron. It steers which trials run; the engine re-weights every trial and a validator strips unsupported figures.

82 % · 52 %

p10–p90 coverage against the model's own priors and against widened priors (target 80 %). Hence the label "model range".

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How to read the evidence markers

Each value carries the kind of evidence behind it, using the markers of the project's research dossier (`docs/PREDICTION-SCIENCE.md`).

`M` Measured in a laboratory, a fleet or an incident report.	`S` Simulated or calculated in the cited source (for example a vendor white paper's model or CFD).	`V` Vendor claim, not independently verified.	`A` Our assumption: no source was found, so the range is deliberately wide.	`calc` Arithmetic done by the team, with the script in `research/scripts/`.

Sources of truth for this document

Every figure traces to a file in the repository: the dossier and the implementation log in `docs/`, `docs/ENGINE_CONTRACT.md`, the engine code in `src/domain/engine/` and `server/director/`, `research/SOURCES.md`, `research/derived/*.json`, `research/data/published-transients/cooling_outage_points.csv` and `reports/validation-2026-11-08-*.md`. Where the dossier and the implementation disagree, this document describes what was **built** and lists the dossier's extras as future work. The charts are generated by `docs/pdf-src/build.mjs`, which reads those files and, for two charts, runs the shipped engine functions (`simulateThermal`, `evidencePosterior`) with the parameters used in the project's own tests. Where a quantity was not measured, the text says "not measured".

SECTION 1

The problem and the claim

When an alarm fires in a data hall, the operator has minutes to answer three questions: is the alarm real, how long until running services are affected, and which response should go first? CenterGuard answers each with a number computed from physics, published reliability data and a seeded statistical engine, and it states how far each number can be trusted.

Why it matters

Outages are expensive. The project dossier verified the Uptime Institute figures and corrected the pitch wording to:

> "In Uptime Institute's 2025 annual survey, 57% of operators said their most recent major outage cost more than $100,000, and 1 in 5 said it cost more than $1 million."
>
> Uptime Institute, *Annual Outage Analysis 2026* (released 13 May 2026). Wording as corrected in PREDICTION-SCIENCE.md §2.9.

These are survey self-reports about each respondent's most recent *major* outage, not a loss model for this facility. A lognormal fit to the two exceedance points gives a median of about $149k and a p90 of about $2.7M `calc`; with only two points behind it, CenterGuard shows it as context next to P(service impact) and never turns it into a single "expected dollars" figure.

What was wrong before the overhaul

A science audit (dossier §1) found that the prototype's numbers looked precise but were not predictions:

Problem found	What it meant	Fixed by
Foreknowledge (audit C1)	Forecasts ran every scripted fault, including future ones: the riser leak that starts at T+4 was "predicted" at T+0 while every reading was normal.	`observable.ts`
Thermal model not physical (S3)	A per-rack curve tuned to the cascade's own assumed delays (circular calibration), although a rack inlet is room air.	`thermalZone.ts`
Overconfident intervals (S1)	Only delay noise was sampled, not parameter uncertainty. p10–p90 contained **24 %** of realised losses against an 80 % target.	`ensemble.ts`
No failure probabilities	Backup equipment never failed on demand, though generators, switches and chillers do, at published rates.	`priors.ts`
Verdicts ignored the error budget (C3)	"Portable ≥ 28 °C" decided the verdict; the installed reading was never compared with the reference.	`evidence.ts`

The claim, stated so it is true of the build

Pitch claim

When a warning comes in, CenterGuard compares the installed sensor with a portable reference reading to get the probability that the condition is real. It then runs thousands of physics-based simulations of each candidate response. Nemotron acts as a red team, steering those simulations toward the compound failures most likely to break the plan, while every reported number comes from the simulator.

The dossier's draft said "tens of thousands". The build runs 2,048 trials per option in round 0 (2 or 3 options) plus at most two stress rounds of 4,096, about 12,000–14,000 trials per evaluation `calc` from `server/director/evaluation.ts`; "thousands" is accurate.

What "science-based" means here

- **Mechanism, not curve fitting.** Heat-up comes from a room energy balance, the model form of Schneider Electric's WP179 (§3.1).
- **Out-of-sample checks.** One unknown is fitted to one published peak; the other cases are predicted and misses reported (§6.1).
- **Published rates for rare events** from the INL/EXT-21-65055 component database (§3.3).
- **Explicit uncertainty.** Parameters without data are sampled widely and labelled assumptions (§7); intervals stay a "model range" until coverage reaches 75–85 % (§6.2).
- **An LLM that cannot move the estimate.** Nemotron changes which trials are drawn, and importance weights correct for that choice (§4).

SECTION 2

System overview

One pipeline turns readings into a recommendation. Each step is a pure, seeded TypeScript module under `src/domain/engine/`, so the same input always gives the same output.

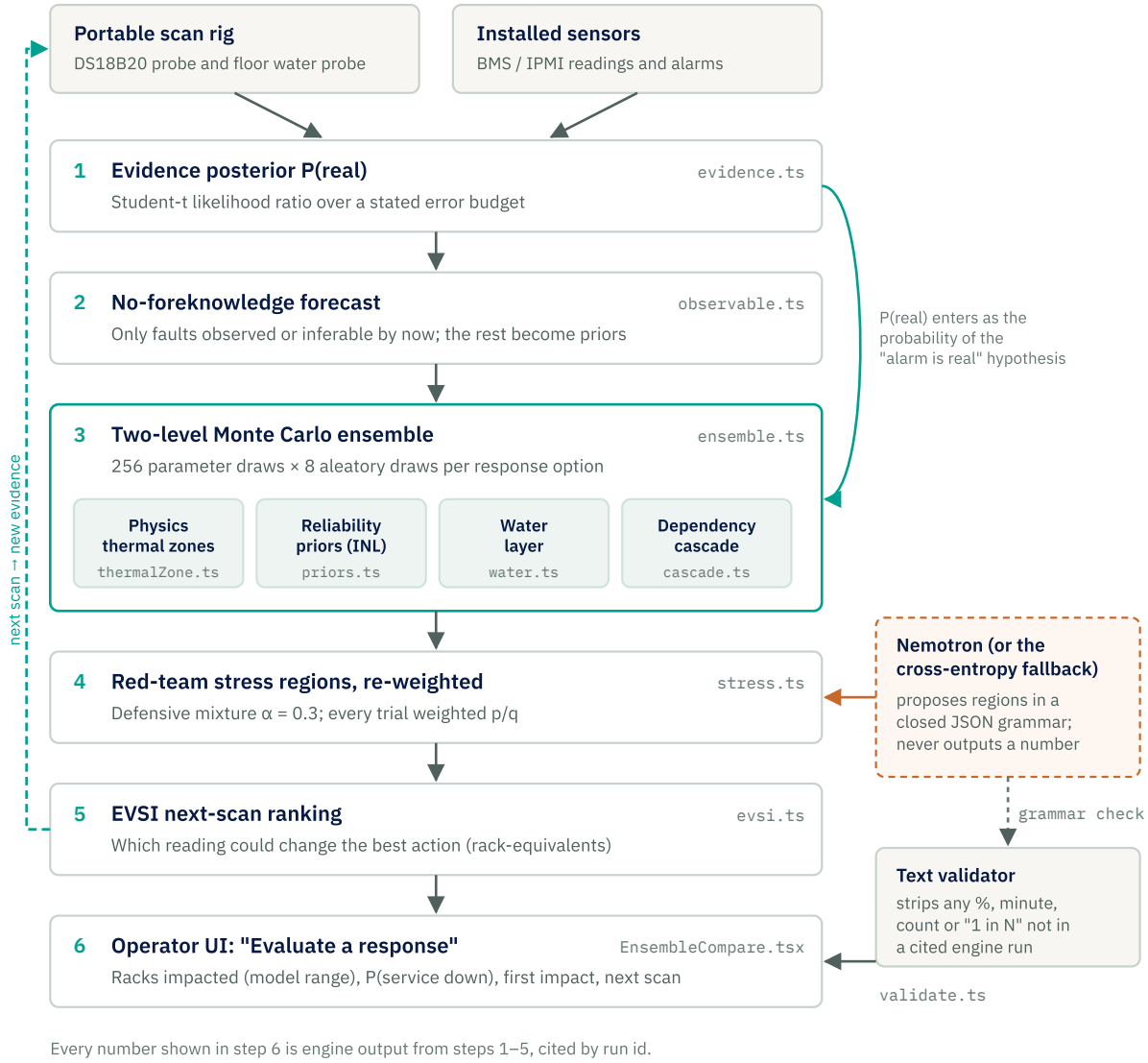


Figure 1. Data flow for one incident evaluation. Solid arrows carry engine data. The orange box is the only place an LLM participates: its proposals pass a grammar check before they can steer sampling, and its prose passes a number validator before the operator sees it. The static demo build runs the same pipeline with the cross-entropy proposer and no network.

Scope note. Forecast mode is opt-in for `simulateScenario` and `predictFix`. The stage, presentation and phone screens still replay the scenario script; only the evaluate path, the director session and the ensemble always forecast (implementation notes, "Decisions").

SECTION 3 · MODELS

The models

Five models produce the numbers: equations in plain language, constants with their evidence markers, and what each model cannot do.

3.1 Heat-up after cooling loss: a room energy balance with two clocks

Idea. When the cooling units stop, the servers keep running on UPS and their heat goes first into the room air, then into the metal of the servers and racks, and, if the fans still run, into the chilled-water loop. A rack has no temperature limit of its own: its inlet *is* the zone's air. CenterGuard therefore models one well-mixed zone per pair of air handlers (CRAH units) with three lumped heat stores, the model form used by Schneider Electric WP179 and its patents US9418179B2 and US10817033.

$$C_a \, dT_a/dt = P_{air} - Q_{coil}(t) - G\,(T_a - T_m) \quad \text{room air (≈ rack inlet)}$$
$$C_m \, dT_m/dt = G\,(T_a - T_m) \quad \text{IT equipment and rack steel}$$
$$C_w \, dT_w/dt = Q_{coil}(t) - Q_{chiller}(t) \quad \text{chilled-water loop}$$
$$C_a = \rho c_p V, \quad C_m = N_{racks}\, m_{rack}\, c_m, \quad G = \boldsymbol{k_G} \cdot \varepsilon \cdot \dot{m}_{IT}\, c_p, \quad Q_{coil} = \text{fans on ? } \dot{m}_{crah}\, c_p (T_a - T_s) : 0$$

C = heat capacity (J/K), T = temperature, P = IT power to air, G = IT–air conductance (W/K), ε = server heat-exchange effectiveness, ṁ = mass flow of air. Implemented in `thermalZone.ts` .

In plain words. Right after the fans stop, only the air absorbs the heat, so it warms fast: for the fixture hall (24 × 24 × 3.5 m, 351 kW to air) the initial slope is P/C_a ≈ 8.7 °C/min `calc` . Within minutes heat flows into the much larger metal mass and the rise slows toward $P/(C_a + C_m)$ ≈ 1.2 °C/min `calc` . The air leads the metal by roughly P/G; that lead, not the rack's own kW, sets how fast 32 °C is reached. With fans off the system has a closed form:

$$T_a(t) = T_0 + P\,t/C + (C_m/C)\,D_\infty\,(1 - e^{-t/\tau}), \quad C = C_a + C_m, \quad \tau = 1 \,/\, [G(1/C_a + 1/C_m)], \quad D_\infty = P\tau/C_a$$

Fans running with chilled water lost (or only coil water left) is integrated with explicit Euler, dt = 2 s. Tests check the closed form against Euler and energy conservation within 0.5 %.

The coupling factor k_G. With textbook coupling (k_G = 1) the model over-predicts the late rise in WP179's simulated room by about 8 °C at 5 min. A multiplier k_G ≈ 2 reproduces the published peaks within about 0.5–3.5 °C (§6.1). Physically it stands for server fans ramping up and for thermal mass the model leaves out (floor tiles, cabling, walls). Because it is fitted to two simulated cases from one vendor, the ensemble samples it as k_G ~ U(1.5, 3) instead of fixing it.

Two clocks, not one "lost" time

The old model called a rack "lost" at 32 °C. That is ASHRAE's A1 *allowable* limit, the edge of the operating envelope, not a failure. Servers warn or throttle higher up: Dell iDRAC warns at 42 °C and goes critical at 47 °C inlet. CenterGuard reports two clocks per zone:

- **Out of envelope** when the inlet crosses **32 °C** (ASHRAE TC9.9 A1 allowable).
- **Service-impact risk** when the inlet crosses **T_{trip} ~ U(40, 45) °C**, anchored on the A3 (40 °C) and A4 (45 °C) allowable limits and the iDRAC thresholds. "Impacted" in every output means this second clock, or a power, water or fire loss.

Constants and priors. Published: ρc_p = 1.19 × 1006 J/m³·K; IT airflow 0.059 m³/s per kW (125 cfm/kW) `S` WP179; IT specific heat 650 J/kg·K (US10817033); service-impact inlet T_{trip} ~ U(40, 45) °C (ASHRAE A3/A4, iDRAC); DLC throttle after CDU flow stops Triangular(0.3, 0.6, 2.0) min, from 23–353 s measured by Alkharabsheh 2017 `M` . Published point, assumed width: rack + IT mass U(350, 700) kg (WP179 uses 545); ε ~ U(0.6, 0.9) (US10817033 gives 0.7–0.9). Fitted: k_G ~ U(1.5, 3) `calc` . Assumed for lack of site data `A` : ceiling U(3.0, 4.5) m, a 0.6 m plenum with p = 0.5, inter-zone mixing U(0, 0.5), loop water log-U(3, 11) L/kW (not identifiable from WP179), coil share U(0.1, 0.3), CRAH airflow 1.2 × IT airflow, coil effectiveness 0.75, DLC share to air 0.15, start temperature 22.5 °C.

How the clocks enter the cascade. For each parameter draw θ the engine computes, once per zone, the minutes from fan loss to each threshold (t_{off}), from loss of chilled-water flow (t_{flow}) and from loss of chilled-water production (t_{chw}). The dependency edges then take their delays from physics: CRAH → rack = t_{off}, pump → CRAH = $t_{flow} - t_{off}$, chiller → pump = $t_{chw} - t_{flow}$, so a chiller loss reaches the racks at t_{chw}. Racks in a zone share one draw, so they heat up together. The four 31 kW CDU-fed GPU racks take the measured-throttle delay instead, counted as their impact time.

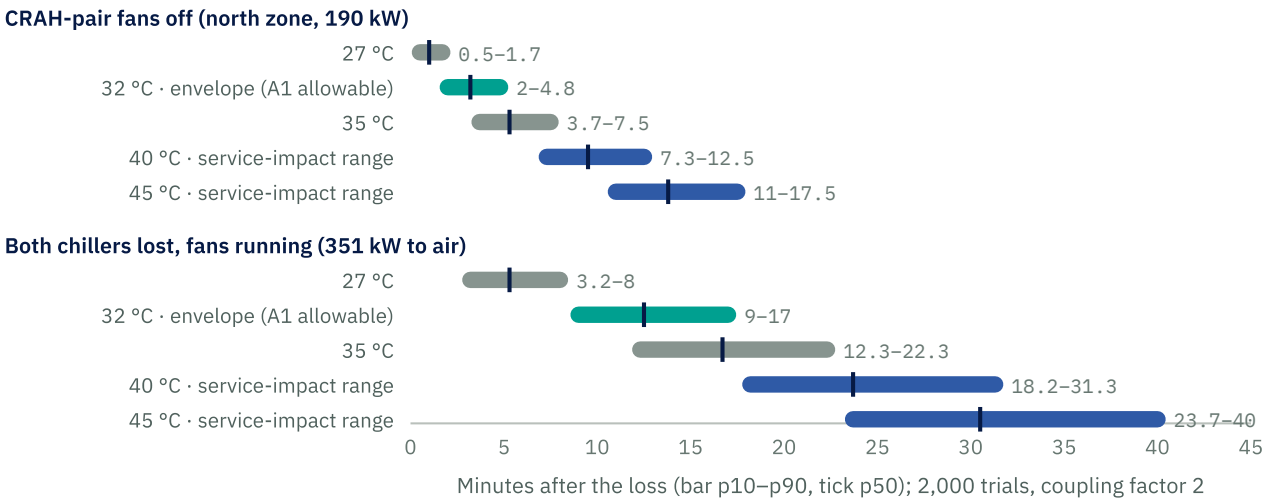


Figure 2. Minutes to each inlet threshold in the fixture hall, from the research Monte Carlo (`research/derived/thermal_backtest_kG2.json`; 2,000 trials over ceiling height, plenum, ε, rack mass, start temperature, loop volume and inter-zone mixing). Losing a CRAH pair's fans is a matter of minutes; losing the chillers while fans run is governed by the loop water and takes several times longer. The pre-overhaul chain (chiller → pump → CRAH → rack, modes 15 + 10 + 6 = 31 min) was too slow for 32 °C and roughly right for 40 °C.

> **What this model cannot do**
>
> It is lumped, not CFD. It cannot represent hot spots, recirculation at a single rack or containment leakage, and one rack's inlet can run several °C hotter than its zone (per-cabinet CFD in Khankari's work reaches 35 °C two to four times faster than well-mixed). Well-mixed is therefore the *slow* bound. The display-only curve `thermal.ts` `inletTempAt` still drives the synthesized readings on screen; predictions use `thermalZone.ts`.

3.2 Is the alarm real? The evidence posterior

Idea. An installed sensor says a rack inlet is hot; a technician holds a portable reference probe at the same point. Instead of a fixed threshold, CenterGuard asks which explanation makes the portable reading more likely under an explicit error budget, and turns that into a probability.

> H_{real}: the installed reading reflects a real condition at the point. H_{meas}: it is a measurement problem (offset, drift, placement, recirculation).
>
> $p(y \mid H_{\text{real}}) = t_{\nu}(\, y \,;\, x_{\text{installed}},\ \sqrt{(\sigma_{\text{inst}}^2 + \sigma_{\text{ref}}^2 + \sigma_{\text{sp}}^2)}\,)$
>
> $p(y \mid H_{\text{meas}}) = t_{\nu}(\, y \,;\, \mu_{\text{baseline}},\ \sqrt{(\sigma_{\text{model}}^2 + \sigma_{\text{ref}}^2 + \sigma_{\text{sp}}^2)}\,)$
>
> $\text{LR} = p(y \mid H_{\text{real}}) \,/\, p(y \mid H_{\text{meas}}), \quad P(\text{real} \mid y) = \text{LR}\cdot\pi \,/\, (\text{LR}\cdot\pi + 1 - \pi)$
>
> y = mean of the portable hold; t_{ν} = Student-t with $\nu = 4$; μ_{baseline} = mean of the neighbours' installed readings (same row, ±2 slots) or the model baseline; π = prior P(real). Implemented in `evidence.ts`.

Why Student-t and not a Gaussian. Heavy tails stop one counterfeit or misplaced probe from producing absurd certainty. Counterfeit DS18B20 probes have been measured with offsets of −3.5 to +3 °C (Petrich `M`); $\nu = 4$ covers that.

ERROR BUDGET AS BUILT (`DEFAULT_ERROR_MODEL`)

Term	1σ / value	Basis
Installed sensor σ_{inst}	1 °C	0.5 (calibrated) to 1.5 °C (drifted), IEC 60751; IPMI bias up to +5 °C, Khalili 2019 `M`; mid value `A`
Reference probe σ_{ref}	0.5 °C	DS18B20 ±0.5 °C (3σ), widened: the scan holds 10 s, the runbook asks 90 s `A`
Placement σ_{sp}	0.5 °C	Same height, 5 cm in front of the inlet: 0.3–0.5 °C (runbook RB-VERIFY-TEMP; Subzero `V`)
Baseline σ_{model}	1.5 °C	Neighbour or zone baseline uncertainty `A`
Prior π	0.5	No base rate of false inlet alarms found `A`; also shown at 0.2 and 0.8
Floor water probe sensitivity / specificity	0.95 / 0.98	`A`
"Unsettled" probe slope	0.2 °C/min	`A`; above it $\sigma_{ref} \geq 1.5 \times$ |slope|, flag `unsettled_lag`; degraded scan quality triples σ_{ref}

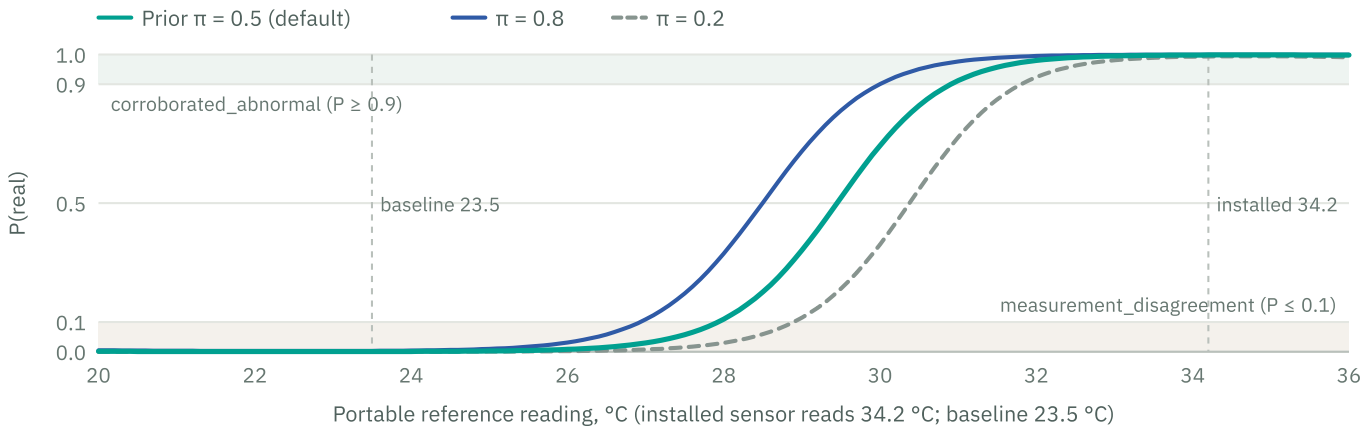


Figure 3. P(real) from the shipped `evidencePosterior` with the default error budget, an installed reading of 34.2 °C and a baseline of 23.5 °C, for three priors. Readings near the baseline favour a measurement problem; readings near the installed value favour a real condition; the middle stays inconclusive and asks for a repeat measurement.

Verdicts. P ≥ 0.9 gives `corroborated_abnormal`, P ≤ 0.1 gives `measurement_disagreement`, anything between is `inconclusive`. The wording describes the measurements, never the equipment: the rationale never says a sensor is "broken" or a rack is "safe". **Measurability gate.** The rig's air probe cannot read chilled-water supply temperature, battery temperature, UPS autonomy, flow or pressure, and its vibration channel is a motion switch, not a mm/s sensor. For those the verdict is `inconclusive` with "not measurable by the portable rig", and P stays at the prior.

Worked example (the dossier's inputs: σ_{inst} 1, σ_{ref} 0.3, σ_{sp} 1.5, baseline 23.5 ± 1.5 °C, installed 34.2 °C). A portable reading of 24.0 °C gives P(real) = 0.0053, so the readings favour a measurement problem. A reading of 31.0 °C gives P(real) = 0.90 (LR ≈ 9.4). The dossier's Gaussian sketch had quoted LR ≈ 110 for the second case; the Student-t likelihood is deliberately more cautious, and the test asserts P > 0.88. For water, a wet probe at π = 0.5 gives P(real) = 0.95 / (0.95 + 0.02) = 0.979.

How P(real) moves the forecast. P(real) becomes the probability of the hypothesis "the alarm is real" in the ensemble (§3.5), so a disagreeing portable reading lowers the predicted impact instead of being ignored. (Probe-settling extrapolation and a slope term in the likelihood are future work, §7.)

3.3 Backup equipment can fail: reliability priors from INL

Idea. A response plan usually depends on backup equipment working when asked: the generator starts, the transfer switch transfers, the standby air handler starts, the chiller restarts. Each of those has a published failure probability per demand. The source is Idaho National Laboratory's *Industry-Average Performance for Components and Initiating Events at U.S. Commercial Nuclear Power Plants: 2020 Update* (INL/EXT-21-65055) `M`, transcribed into `research/derived/reliability_priors.json`.

How they are sampled. Each rate is a Beta(f + 0.5, n − f + 0.5) posterior (Jeffreys prior) for *f* failures in *n* demands. For one demand, a Bernoulli trial with Beta-distributed p is exactly Bernoulli(Beta mean), which is what the engine samples.

Sampled per demand in the engine (`priors.ts`): generator fails to start 2.22e−3 (INL EDG-FTS, Beta(136.5, 61227.5)); ATS fails to transfer 1.33e−3 (INL ABT-FF); UPS drops load on transfer 1.0e−3 (Schneider WP75 `V`); standby CRAH fails to start 2.71e−3 (INL AHU-FTS-NS 2.11e−4 plus 2.5e−3 maintenance unavailability); chiller fails to restart 2.51e−2 (INL CHL-FTS-NR 5.09e−3 plus 2.0e−2

unavailable).

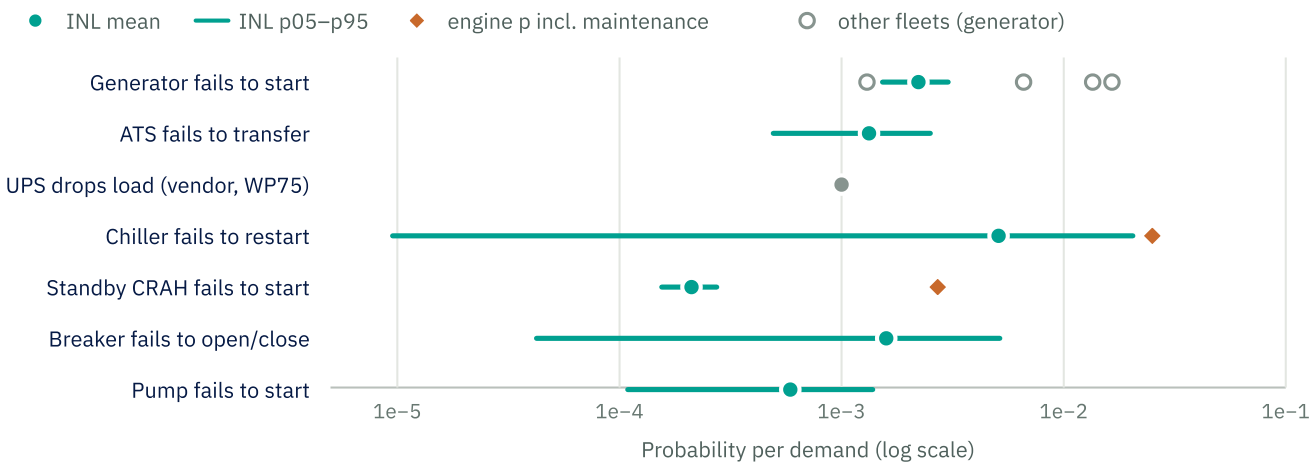


Figure 4. Per-demand failure probabilities (`reliability_priors.json`). Teal: INL mean and 5–95 % range. Orange: the value the engine samples where maintenance unavailability is added. Hollow gray: generator fail-to-start in other fleets (Navy, NRC fleet via NREL, IEEE 493 via WP75, a poorly maintained Hong Kong set at 1.65e−2). Breaker and pump rows are listed for context; they are not ensemble families.

Unobserved future faults as hypotheses. A scripted fault that nobody could see yet becomes a hypothesis whose prior is the chance of failing during the remaining horizon, 1 − exp(−rate × hours). The per-hour rates are INL means: chiller fail-to-run 3.87e−5/h, air handler 2.61e−6/h, pump 2.26e−6/h, inverter (UPS) 3.49e−6/h, battery 4.05e−7/h, breaker spurious opening 1.73e−7/h, ATS spurious 1.25e−7/h, generator fail-to-run while running 3.31e−3/h, manual-valve small external leak 8.67e−8/h. For the chiller incident, "CH-2 also trips" over the remaining 58 minutes therefore has a prior of about 3.7e−5 `calc`. If a caution-level precursor is already active (CRAH-2 vibration), P(trip within the horizon) = 0.02, range 0.005–0.1 `A`: no P-F interval data were found.

Why rare events need care. At 2.2e−3, a 400-run Monte Carlo sees on average about 0.8 generator failures. Plain sampling cannot estimate those tails well, which is the reason for the importance sampler of §4.

Caveat. Nuclear-plant equipment is heavily tested and maintained, so these rates are optimistic for commercial sites. The 1.65e−2 Hong Kong figure is the "poorly maintained" bound; it is recorded in the JSON but has not yet been run as a sensitivity case (not measured).

3.4 Water: the leak model

Idea. A chilled-water leak spreads across the floor and fails whatever it reaches (UPS, batteries, PDUs). The water layer (`water.ts`) is a height-field model on a 0.5 m grid: each step, every cell passes water to lower neighbours by surface height (ground + depth), mass-conserving; equipment footprints are 5 cm plinths; zones carry their own floor elevation; a source adds the leak rate in L/min; an asset is "wet" when water under its footprint reaches 1 mm. It is not a fluid solver (no momentum, no raised-floor plenum flow), and the UI keeps the "approximation" label.

> Puddle thickness on a flat floor: $h = 2\,l_c \sin(\theta/2)$, $l_c = \sqrt{\gamma / \rho g} = 2.7$ mm $\rightarrow$ $h = 1.4$–4.2 mm for contact angles $\theta = 30$–$100°$
> Orifice check: $Q = C_d\,A\,\sqrt{(2\Delta P/\rho)}$, $C_d = 0.61$ $\rightarrow$ a 50 mm² joint gap at 3–8 bar leaks 45–73 L/min
> Capillary physics after de Gennes, Brochard-Wyart and Quéré (2004); orifice table in dossier §2.3 `calc`.

The fixed 4 mm film in `water.ts` is therefore the sealed-floor upper end, and the scenario's 38 L/min is consistent with a joint failure. In the ensemble, the **rate** is *observed* as the make-up-water flow mismatch, so only the meter error is sampled: multiplier LogNormal(median 1, σ_{ln} 0.15) `A`. The **film** is U(1.5, 4.0) mm from the capillary formula. The **start** is unobserved: time inferred − U(0, 10) min `A`. **Arrival** times come from cached water-layer runs (no isolation, or isolation after 3, 6, 10, 15, 22 or 30 min), scaled by (film / 4 mm) / rate multiplier, a volume-limited front `A`; closing the valve stops the water at the fix time.

No foreknowledge in action. In the riser-leak scenario the leak starts at T+4 but nothing observes it. It becomes inferable at T+11, when the make-up-water flow mismatch appears, and is evaluated at T+12. A forecast at T+0 or T+4 contains no leak clocks; the tests assert this, and that a forecast is unchanged when unobserved future faults are added or removed.

3.5 The cascade and the two-level ensemble

The cascade. The facility is a dependency graph: utility → generator → ATS → UPS → PDU → rack for power, chiller → pump → CRAH or CDU → rack for cooling, and derived "shares space" edges for water and fire. Each edge has a delay. A node's loss time is the minimum over its redundancy groups of the maximum over that group's members:

$t_{\text{loss}}(n) = \min_{\text{groups } g} \max_{m \in g} [\, t_{\text{loss}}(m) + \text{delay}(m \to n) \,]$

AND within a redundancy group (a CRAH pair fails the rack only when both units are gone), MIN across groups (any one independent path is enough). Propagated by a Dijkstra-like relaxation; fixes recover nodes from their effect time. `cascade.ts` was split into `compileGraph` + `runOnce`, checked bit-identical with the old `runCascade` on six scenarios.

Two levels of uncertainty. The old engine sampled only the edge delays, independently on every edge, so the 44 rack edges of one room averaged out and looked certain. The ensemble separates what we do not *know* (epistemic, shared by everything in a trial) from what *varies* trial to trial (aleatory):

Outer (256 draws): $\theta \sim q(\theta)$: physics parameters, on-demand failures, battery health, crew speed, leak film/rate/start, hot-spot growth, hazard spread, hypotheses. weight $w(\theta) = p(\theta)/q(\theta)$

Inner (8 draws per θ): the remaining triangular edge delays, generator start-to-transfer, DLC throttle time, battery runtime at load, which UPS drops on transfer

$P_{\text{est}}(F) = \Sigma_i\, w_i\, F_i / \Sigma_i\, w_i$, ESS $= (\Sigma w)^2 / \Sigma w^2$, 90 % interval from the outer-level (cluster) variance; no events → [0, 2.3/ESS]

F = any criticality ≥ 4 service with every replica impacted within the 60-minute incident horizon. Without a stress mixture q = p, except that hypotheses with p outside 0.3–0.7 are drawn at 0.5 and weighted p/0.5, so both branches are always explored. `ensemble.ts`.

Ingredients. Cooling edges take their delays from the zone model (§3.1). Generator → ATS is Triangular(0.15, 0.5, 3.2) min, spanning NFPA 110's 10 s, WP179's 60 s and WP75's 190 s. On a utility loss, generator fail-to-start, ATS fail-to-transfer and a UPS drop become faults at their §3.3 probabilities (which UPS drops is 50/50 A). Battery ride-through is the fixture triangle × state of health, string A U(0.9, 1.0) and string B, 11 years into a 10-year life, U(0.6, 0.85) A. A fix takes effect at t + fixture delay × a crew multiplier LogNormal(median 1, error factor 2) A (no data-center response data were found); a chiller restart at t + 3 × crew + U(2.5, 15) min; standby CRAH, chiller restart and manual generator start can fail on demand. In the rack-inlet incident, a real hot spot relaxes toward now + U(0, 12) °C with time constant U(5, 20) min; shedding load halves the excess and portable cooling cuts it to 30 % A. Fire spread speed carries a LogNormal(1, $\sigma_{\ln}$ 0.3) multiplier A.

What each response option gets. Racks impacted at p10/p50/p90 (weighted quantiles) and mean; P(critical service down) with its 90 % interval; minutes to the first service impact and to the first 32 °C crossing; P(impact) per rack; and an expected utility $U = -\Sigma_{\text{racks}}\, w_{\text{crit}} \cdot 1[\text{impacted}] - \lambda \cdot \text{crew-minutes}$, with w_{crit} the highest service criticality on the rack and $\lambda = 0.05$ rack-equivalents per crew-minute A, shown in the UI. The seed is derived from the incident, the time and the evidence, so a replay is identical.

Incident scenarios: evaluating each warning on its own

Each demo warning is evaluated in a scenario built from the injected incident alone (`src/domain/scenarios/incident.ts`), with no scripted future:

Warning	Hypotheses sampled	Options compared
A-11 inlet 34.2 °C	Real local hot spot, at P(real)	No response · shed load · portable cooling
CH-1 supply alarm	CH-1 really stopped, at P(real); CH-2 also trips, at the INL rate	No response · restart CH-1
PIPE-L2 make-up-flow mismatch	The leak is real, at P(real); start unknown	No response · isolate valve · transfer UPS-B
CRAH-2 vibration 6.8 mm/s	Fan trips within the hour, at P(real) × 0.02 A	No response · swap to standby CRAH

SECTION 4

How Nemotron is used, and why it cannot bias the numbers

Nemotron (`nvidia/nemotron-3.5-lightning-30b-a3b`) works as a red team. It proposes compound failures that might break the chosen response, and the engine tests them. It never produces a reported number. Its proposals change which trials are drawn; importance weights then correct for that choice, so a bad idea costs precision, never correctness.

The loop

1. **Round 0** (`stress_digest`, cited as `stress-1`). The ensemble runs every response option (2,048 trials each) and builds a *failure digest*: the families over-represented among failing trials (lift = P(top or bottom decile | fail) / 0.1; for Bernoulli families P(fails | fail) / P(fails)), the ESS and the estimate.

2. **Proposals** (`stress_test`): up to 6 regions, each a label, a rationale and up to 6 terms. `validateProposal` checks every term against the grammar and the family list; invalid terms go back to the model with a reason.
3. **Re-weighted sampling.** The engine draws 4,096 trials (512 × 8) from the defensive mixture and attributes the failure probability to the regions.
4. **At most two stress rounds**, then a recommendation that passes the text validator. Without Nemotron (no key, or the static artifact) the cross-entropy method proposes the regions, with the same output schema.

The closed grammar

Operation	Allowed on	Constraint
`quantile_window`	continuous families	0 ≤ lo < hi ≤ 1 and width ≥ 0.01: sample the family only from that slice of its prior's quantiles
`force`	Bernoulli families	value 1 ("fail") or 0 ("ok")
`set_p`	Bernoulli families	0 < p ≤ 0.5
`scale`	log-scale families (lognormal, log-uniform)	f ∈ [1/4, 4]

Families are the engine's own (for example `gen_fts`, `chiller_restart_min`, `battery_soh_B`, `k_G`, `crew_mult`, `h:real`). A family may appear once per region. Anything else (an unknown family, a window on a Bernoulli, set_p above 0.5, a seventh term) is rejected. Every allowed operation has a closed-form density, which is what makes exact re-weighting possible.

Why the estimate stays honest

$$q(\theta) = \alpha \cdot b(\theta) + (1-\alpha)\cdot(1/K)\,\Sigma_k\, q_k(\theta), \quad \boldsymbol{\alpha = 0.3}$$

$$w(\theta) = p(\theta)\,/\,q(\theta) = [p(\theta)/b(\theta)]\cdot[b(\theta)/q(\theta)], \quad b/q = 1\,/\,(\alpha + (1-\alpha)\cdot\mathrm{mean}_k\, q_k/b) \le 1/\alpha \approx 3.3$$

$$P_{\mathrm{est}}(F) = \Sigma\, w\, F\,/\,\Sigma\, w \quad \text{(self-normalised; consistent for any valid } q_k)$$

b = the base sampler (the prior, with hypotheses stratified at 0.5); q_k = region k's proposal. `stress.ts`, `ensemble.ts`.

- **Defensive mixture.** 30 % of draws always come from the base sampler, so every part of the prior keeps being sampled, whatever Nemotron proposes.
- **Weight cap.** Relative to the base sampler, no trial can weigh more than 1/α ≈ 3.3. By Hesterberg (1995), a useless proposal therefore costs at most about a 3.3-fold increase in variance over plain Monte Carlo; a good one cuts the variance of a rare-failure estimate. `stressTest` reports the largest weight seen so the bound is checked, not assumed.
- **Attribution by the engine.** For each region the engine computes its prior mass ("about 1 in N incidents"), P(failure | region) and its share of the total failure probability. Nemotron reads these; it does not compute them.
- **Stop rule.** A stress result is marked converged when ESS > 2,000 and the 90 % half-width is below 25 % of P_{est}.

Tested. `tests/engine/stress.test.ts` checks that weights stay within 1/α of the base sampler, that a generator or ATS failure is recovered within 3 standard errors by importance sampling with N = 8,000, and that random valid proposals average to the true value (no bias).

The text validator: Nemotron's prose cannot carry a new number

`server/director/validate.ts` runs on every recommendation. Each engine run's numbers are registered under its run id (`run-N`, `stress-N`), in raw, rounded and percent forms. The validator then scans the summary, next action, next-scan reason and every hypothesis:

Check	Rule
Clocks	"T−14", "T+44", "14 min", "12–20 min" must match a cited run's number within ±1 (rounding), or a number quoted from the inputs
Percentages	"3.2 %" must match within max(0.15, 5 %) of a cited or observed value
Counts and odds	"12 racks", "2 services" must match exactly; "1 in 400" within max(1, 5 %)
Run ids	Unknown run ids are removed and reported
Direction	"later", "more time" or "buys" next to a cited shift whose sign is negative (or the reverse) is replaced by the engine's own templated sentence
Provenance	Claims that a rig is trustworthy, a sensor is broken or faulty, or equipment is safe are removed

Anything that fails becomes `[unsupported]` and is logged as a warning. The director also has a 30 s wall-clock cap and falls back to the local planner.

What happened in the live runs

Evaluation	Wall time	Requests	Prompt tokens	Outcome
Heat (CH-1), P(real) 0.5	16.0 s	12	86k	2 regions proposed (chiller restart failure via set_p; generator fail-to-start); 0 validator warnings
Water (PIPE-L2), P(real) 0.98	18.4 s	12	93k	5 proposals rejected by the grammar (force on a continuous family; a 0–30 window) before 2 valid rounds

The rejections are the grammar working as designed: invalid proposals never reached the sampler. After this run the digest and prompt were changed to list families by type and ask for narrow tails. Whether Nemotron's regions reduce variance more than the cross-entropy fallback at equal trial counts is **not measured** yet; until it is, the honest claim is that Nemotron matches the fallback's role and explains the failure modes in operator language.

Choosing the next scan: expected value of sample information

The scan suggestion used to come from an ad-hoc score. It now comes from a preposterior decision analysis (Raiffa and Schlaifer 1961), computed from the ensemble's own trials with no new simulation:

> $\mathrm{EVSI}(s) = \Sigma_y \max_a \Sigma_i w_i L_y(i)\, U_a(i) / \Sigma w \;-\; \max_a \Sigma_i w_i U_a(i) / \Sigma w$
>
> $L_y(i)$ = probability of scan outcome y in trial i: a floor probe reads wet with sensitivity 0.95 where that trial's water has arrived, else with 1 − specificity; an inlet probe reads near the installed value where the hot spot is real, else near the baseline (Student-t budget of §3.2, five bins). U as in §3.5. Units: rack-equivalents. `evsi.ts` .

EVSI is never negative and is exactly zero when no outcome would change the best action. Only scans the rig can physically make are candidates. Tests check that the A-11 inlet scan ranks first for the A-11 alarm, that a floor scan outranks an inlet scan for the riser leak, and that EVSI matches a brute-force two-hypothesis calculation on a toy ensemble (0.265). **An honest finding:** at the riser leak at T+12, isolating the valve is the best action whatever a single floor point reads, so every EVSI is 0. The ranking then falls back to how far the reading would move P(service down) (`forecastShift`), and the UI says so.

SECTION 5

Data pulled

About 30 MB in 9 files across 7 source folders, retrieved on 2026-11-07, with URL, date and licence for each in `research/SOURCES.md` . All downloads are treated as untrusted data: nothing downloaded is executed, and the analysis scripts only read the files as text.

Dataset	Volume	What it was used for in the build
Open-Meteo archive, DFW airport grid cell, hourly 2016–2025 (ERA5 reanalysis; CC BY 4.0)	87,672 hours × 5 variables (dry-bulb, RH, dew point, wet-bulb, pressure), 3.4 MB	Heat-wave exposure and chiller-derate hours (Figure 5). *Context only:* the planned ambient prior for the heat-wave scenario is not yet sampled by the ensemble.
NOAA NCEI Global Hourly, station 72259003927 (DFW), 2023, 2024, 2025 to 27 Aug	23,165 routine hourly reports, 20 MB raw CSV	Ground truth for the reanalysis: daily-maximum bias −0.2 °C (5th–95th percentile −2.6 to +2.5 °C, 970 days)
INL/EXT-21-65055 component reliability, 2020 update	4.4 MB PDF, about 100 component and failure-mode rows used	Every on-demand failure probability and per-hour hypothesis rate (§3.3), via `reliability_priors.json`
ASHRAE TC9.9 5th-edition reference card	160 KB PDF	The 32 °C envelope clock; the 40 / 45 °C anchors of the service-impact clock
Schneider Electric WP179	616 KB PDF	Energy-balance model form; the backtest targets (§6.1)
Tradat et al. 2018, Binghamton (measured)	1.0 MB PDF	Measured ride-through with fans running: 74 kW in a 215 m² lab reached 32 °C in 22–46 min (IPMI vs external sensors); server IPMI sensors read hot and under-estimated ride-through by 12–42 %

Dataset	Volume	What it was used for in the build
Published transients, hand-transcribed	30 points from 6 sources (WP179 and its CFD case, Opengate 2007, Khankari, Tradat 2018, Alkharabsheh 2017, Intel 2007)	Backtest targets in `cooling_outage_points.csv`; each row carries its URL and an M / S / V marker

Read online but not downloaded: Alkharabsheh 2017 (DLC throttle times), Khalili 2019 (server-inlet sensor bias), Petrich (counterfeit DS18B20), Schneider WP75 (UPS and generator inputs), NREL generator reliability, AMCA 204-20, the Uptime Institute releases, the Google Cloud europe-west9 incident report, Trane and Daikin restart pages. Derived outputs: `dfw_weather_summary.json`, `thermal_backtest_kG1.json`, `thermal_backtest_kG2.json`, `reliability_priors.json`.

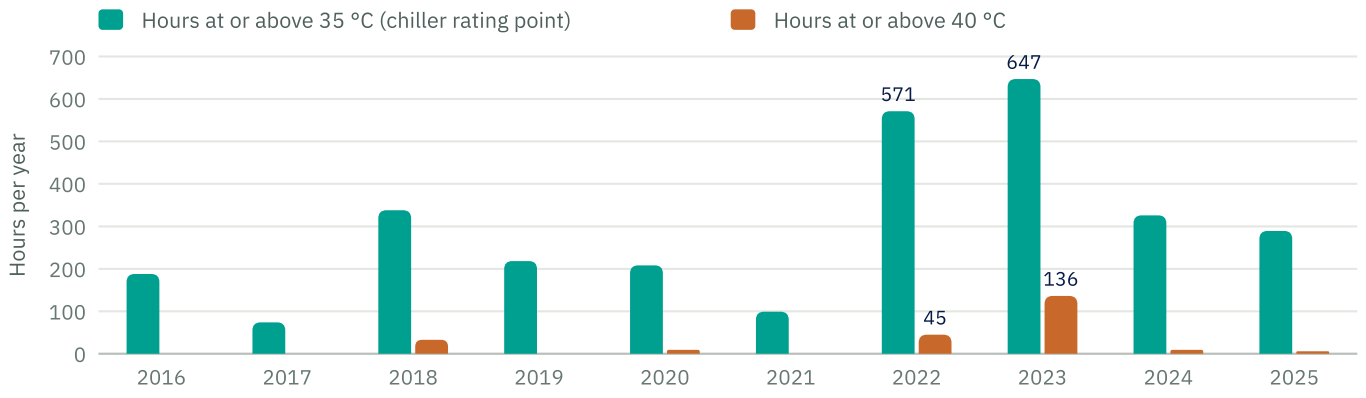


Figure 5. Hours per year at or above 35 °C and 40 °C dry-bulb in Dallas–Fort Worth (Open-Meteo, `dfw_weather_summary.json`). Air-cooled chillers are rated at 35 °C and lose capacity above it. 2023 was the extreme year: maximum 43.3 °C, 99.6 % design value 41.6 °C, 136 hours at or above 40 °C (the DFW station recorded 124). With an assumed derate of 2 %/°C, 2023 had 130 hours below 90 % chiller capacity. February 2021 (Winter Storm Uri) had 50 hours at or below −10 °C, minimum −17.8 °C; cold-weather generator effects are not modelled.

What could not be obtained

- IEEE 493 Annex Q tables and IEEE 3006.8 (paywalled): the Gold Book generator figure is used only through WP75's citation.
- Manufacturer chiller capacity-versus-ambient tables (they live in selection software), so the derate per °C is an assumption, U(1.0, 2.0) %/°C `A`.
- Any measured data-center leak-spread data; any base rate of false temperature or leak alarms; full texts of Zhang and VanGilder 2011 and Lin et al. 2014.
- No local PDF renderer was available, so PDF numbers were read through a web-fetch tool. **WP179 Table 2 was reconstructed from scrambled PDF text** and must be checked against the PDF before it is quoted; this document quotes no value from that table.

Licences: Open-Meteo CC BY 4.0 ("Weather data by Open-Meteo.com"); NOAA US-government data, no restrictions; INL publicly released via OSTI; ASHRAE, Schneider Electric and ASME PDFs are reference copies only and must not be redistributed (keep `research/data/**/*.pdf` out of any public repository).

SECTION 6

Validation, reported honestly

Two kinds of check: the physics against published cooling-outage transients, and the statistics against a harness that knows the "true" outcome. Both are reported with their misses. Neither is yet a replay of a measured site incident, which is the next real validation.

6.1 Thermal backtest against WP179 and its CFD room

Procedure. Configure the model with WP179 Example 1's stated room (30 × 24 × 4 m plus a 0.6 m plenum, 400 kW, 100 racks of 545 kg, generator at 60 s, chiller restart at 12 min). Fit only the one declared unknown, the loop water volume, to the published 41.1 °C baseline peak. Then *predict* the 5-minute and 4-minute restart cases, the recovery time and the CFD room. The curves below come

from running the shipped `simulateThermal` with the exact parameters of `tests/engine/thermalZone.test.ts` (k_G = 2, ε = 0.8, instant chiller restart).

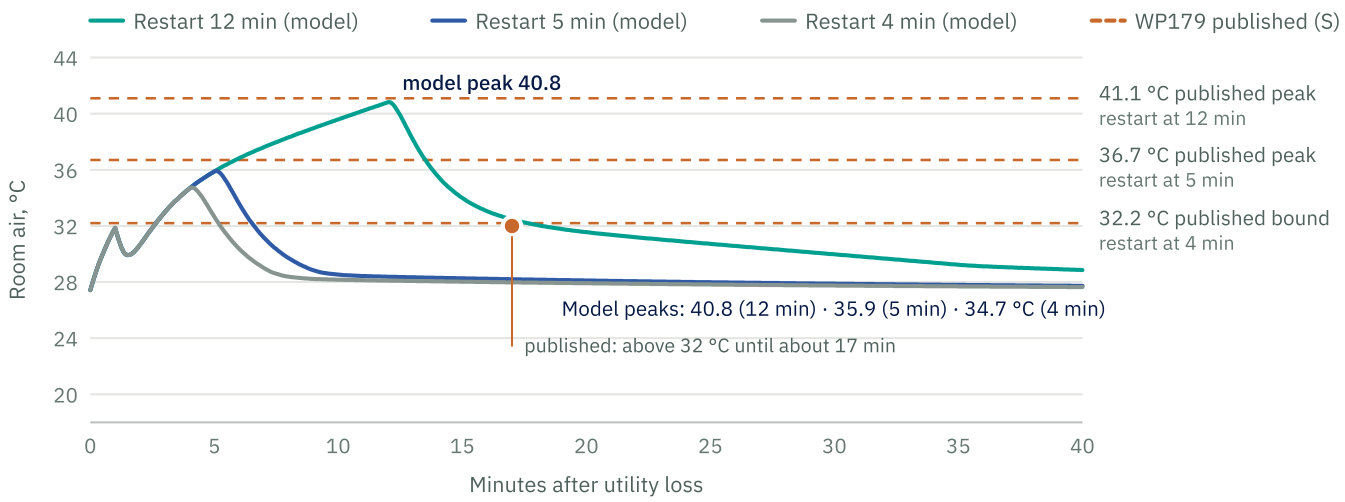


Figure 6. WP179 Example 1. Model curves (solid) against the published peaks (dashed) and the published "unacceptably high until about 17 minutes" point. The 12-minute baseline is the fitted case; the 5-minute restart is predicted within 1 °C; the 4-minute restart is over-predicted, a known miss.

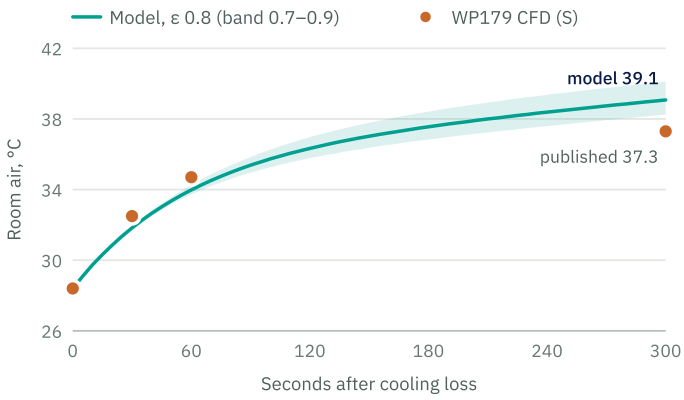


Figure 7. WP179 CFD room with no cooling recovery. The IT load is illegible in the source, so it was inferred from the 0–30 s slope (59 kW) and 15 racks were assumed: a consistency check, not an independent validation.

Reading the backtest.

- Textbook coupling (k_G = 1) over-predicts the late rise badly: 45.6 °C against 37.3 °C at 300 s in the CFD room.
- k_G = 2 meets the CFD criteria and the Example 1 baseline and 5-minute peaks.
- It misses the 4-minute peak by about 2.5–3.5 °C. A one-node water loop cannot reproduce how fast a restart recovers; a two-node loop is the likely fix and is future work.
- The fitted loop volume falls to the grid floor at k_G = 2: 0.25 m³ (0.6 L/kW) at k_G = 2, against 1 m³ (2.5 L/kW) at k_G = 1. The loop volume is therefore **not identifiable** from WP179 and is sampled from practice instead (§3.1).

Published target (WP179, S)	Published	k_G = 1	k_G = 2	Acceptance	k_G = 2 vs acceptance
Ex. 1 baseline peak, restart at 12 min (fitted)	41.1 °C	41.7 / 41.1	40.8 / 41.1	± 2.5 °C	fitted
Ex. 1 peak, restart at 5 min	36.7 °C	32.6 / 33.6	35.9 / 36.5	± 2.5 °C	within (−0.8 / −0.2)
Ex. 1 peak, restart at 4 min (published as a bound)	32.2 °C	32.6 / 32.6	34.7 / 35.6	± 2.5 °C	partly within (+2.5 / +3.4)
Ex. 1 last minute above 32 °C	17.0 min	20.8 / 25.3	18.2 / 26.4	± 5 min	partly within (+1.2 / +9.4)
CFD room at 60 s	34.7 °C	35.2	34.0	± 1.5 °C	within (−0.7)
CFD room at 300 s	37.3 °C	45.6	39.1	± 3 °C	within (+1.8)

Model columns give an instant chiller restart / a 4-minute capacity ramp (CFD rows have no restart). Acceptance limits are the dossier's §7.1 criteria. Values from `research/derived/thermal_backtest_kG1.json` and `_kG2.json`; the shipped port reproduces the k_G = 2 instant-restart column (40.8, 35.9, 34.7 °C; CFD 34.0 and 39.1 °C). **Caveat on WP179:** its values were read through a web-fetch tool, not a PDF renderer, and transcribed by hand from figure text (106 °F, 98 °F, "only marginally above" 90 °F); they should be re-checked against the PDF. No value from WP179 Table 2 is used. Not run as tests: the Tradat 2018 fans-running shape check and the Alkharabsheh 2017 throttle-time factor check (not measured); the DLC prior is taken directly from the measured range.

6.2 Are the intervals honest? Coverage

Harness. `npm run validate` runs 36 scenarios (6 named, 3 of them modelled on real incidents, plus 30 random) × 50 iterations, seed 1. Each iteration draws a "reality" and asks whether each realised loss time falls inside the predicted p10–p90. For an honest 80 % interval the answer should be yes about 80 % of the time. The default perturbation widens delays ±30 % with a per-iteration bias of ±20 %, shifts start times ±2 min, scales leak and spread rates ±40 %, and misses a first signal 20 % of the time (up to 5 min of extra detection lag).

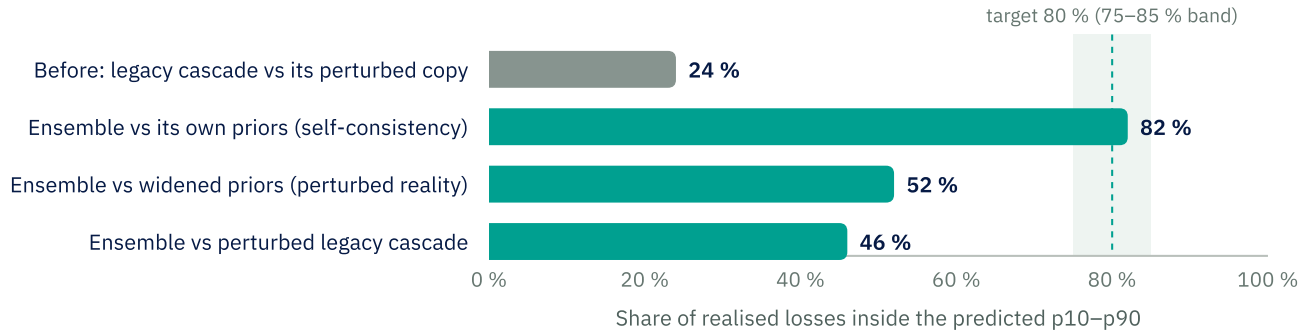


Figure 8. p10–p90 coverage from `reports/validation-2026-11-08-*.md`. Target 80 %, acceptable band 75–85 %.

Model	"Reality" it is judged against	Coverage	p50 |error|	Bias	False alarms	First rack right
Legacy cascade (before)	Perturbed copy of itself	**24 %**	3.0 min	+0.9 min	10 %	53 %
Ensemble	Draws from its own priors (self-consistency)	**82 %**	3.0 min	+0.5 min	0 %	85 %
Ensemble	Priors widened ±30 %, starts ±2 min, rates ±40 %	**52 %**	4.9 min	+0.8 min	1 %	81 %
Ensemble	Perturbed legacy cascade	**46 %**	5.9 min	+4.3 min	10 %	53 %

What the three numbers mean.

- **24 % before.** The legacy model sampled only delay noise, so its intervals were far too narrow. By asset type, chiller coverage was 4 % and ATS 5 %; under the perturbed reality the ensemble lifts every asset type, to between 38 % (chiller) and 61 % (battery).
- **82 % self-consistent.** When reality is drawn from the same priors, the ensemble's intervals contain 82 % of outcomes, inside the 80 ± 3 % target. This is a correctness check of the weighting and the quantiles (a bug would show here), not a claim about the real world.
- **52 % under widened priors.** When reality departs from the priors (wider delays, shifted starts, different rates), coverage drops to 52 %, below the 75–85 % band. Coverage more than doubles, but the intervals are still too narrow when the priors are wrong, and the median error grows from 3.0 to 4.9 minutes.

> **Why the UI says "model range"**
>
> Until validated coverage lands in 75–85 % under the perturbed reality, the interval is not called a p10–p90. `coverage.ts` generates the UI note from the latest results: *"Model range, not a calibrated p10–p90: validated coverage 52 % against the 80 % target when reality departs from the priors (82 % when it does not; validation 2026-11-08)."*

6.3 First-rack accuracy

"First rack right" asks whether the model names the rack that is actually impacted first. It rose from **53 %** of 1,766 judged iterations (legacy) to **81 %** of 1,759 under the perturbed reality and **85 %** of 1,796 under self-consistency. Predicted versus realised racks saved by a fix was checked only for the legacy model, which predicted 32.1 racks saved on average against 23.6 realised, an overstatement of about 36 %. The ensemble runs did not check fixes ("0 scenarios checked"), so fix accuracy for the new engine is **not measured**.

6.4 Engine tests that guard the science

Prior-predictive coverage within 72–88 %; evidence-posterior calibration on synthetic truth (each P(real) bin within ±0.05); no foreknowledge (forecasts invariant to unobserved future faults, a property test); bounded importance weights, a rare event recovered within 3 SE at N = 8,000 and unbiased random proposals; EVSI ≥ 0 and equal to brute force on a toy case; the thermal backtest, energy conservation and closed form against Euler. 282 tests passed at the end of the overhaul.

6.5 Runtimes

Operation	Measured	Where
Ensemble, 2 options × 2,048 trials, cold (includes the water-layer cache)	0.49 s	Node
Ensemble, 4 options, warm	0.17 s	Node
One cascade run (pre-overhaul engine bench: 400 runs in 14 ms)	22–36 µs	Node, `bench_engine.ts`
Full evaluation with the local planner (CE fallback)	0.15–0.7 s	Node (test bound 3 s)
Full evaluation with live Nemotron (heat / water)	16.0 s / 18.4 s	bridge server, 12 requests each
Validation harness, 36 × 50: legacy / prior-predictive / perturbed / vs legacy	51 / 6 / 207 / 25 s	Node, report headers
Ensemble in the browser (static artifact)	not measured	checked to run, not timed

SECTION 7

Limitations and assumptions

The physics removes the structural error of the old model (wrong shape, circular calibration). It does not remove parameter error: the fixture hall has no measured ceiling height, raised floor, loop volume, CRAH capacity or battery curve. Every value below has no direct source and is sampled or fixed as an explicit assumption.

Assumption in the build	Value
Thermal zone model	
Ceiling height	U(3.0, 4.5) m
Raised-floor plenum present	0.6 m with p = 0.5
Air exchanged with the running half of the hall	U(0, 0.5)
Chilled-water loop volume (3–10 gal/ton practice, not verified)	log-U(3, 11) L/kW
Share of loop water left in coils when pumps stop	U(0.1, 0.3)
CRAH airflow; coil effectiveness; loop water start temperature	1.2 × IT airflow; 0.75; 10 °C
DLC racks' share of power to room air	0.15
Zone start temperature	22.5 °C
Rack mass and ε ranges (point values are published, the widths are not)	U(350, 700) kg; U(0.6, 0.9)
Power path	
Battery state of health, string A / string B	U(0.9, 1.0) / U(0.6, 0.85)
Battery runtime at load (no vendor curve entered)	fixture triangles × SoH
Which UPS drops load on a transfer failure	50 / 50
Delays of edges not driven by physics (fixture values)	fixture triangles
Response and people	
Crew task-time multiplier (no data-center response-time data found)	LogN(median 1, error factor 2)
Base fix delays (isolate valve 4, standby CRAH 10, chiller reset 3 min before the restart ramp …)	fixture values
Chiller restart to capacity, chiller type unknown	U(2.5, 15) min
Utility weight of a crew-minute	λ = 0.05 rack-equivalents

Assumption in the build	Value
Evidence posterior	
Installed-sensor σ (mid of a published range)	1.0 °C
Reference σ widened for the 10 s hold	0.5 °C
Baseline σ	1.5 °C
Prior P(alarm is real) (no base-rate data)	0.5; shown at 0.2 / 0.8
Floor probe sensitivity / specificity	0.95 / 0.98
Slope above which the probe is "unsettled"	0.2 °C/min
Water	
Drain removal (no source)	30 % of local depth per min
Make-up-flow meter error on the observed rate	LogN(1, σ_{ln} 0.15)
Leak start before detection	U(0, 10) min
Arrival scales with film / rate (volume-limited front)	arrival ∝ film / rate
Equipment plinth height	5 cm
Hypotheses and observation	
P(trip within horizon | caution-level precursor) (no P-F interval data)	0.02 (0.005–0.1)
Asset types without a published per-hour rate	1e−5 /h
CDU failure rate taken from the pump rate	2.26e−6 /h
A fault with no observation step is alarmed when it happens (BMS trip)	observed at t_{fault}
A fire is observed when it starts (smoke detection)	observed at start
Local hazards	
Hot spot further rise; time constant; effect of shed load / portable cooling	U(0, 12) °C; U(5, 20) min; 0.5 / 0.3
Fire spread speed multiplier	LogN(1, σ_{ln} 0.3)
Not yet in the build	
Chiller derate above 35 °C (manufacturer tables not found)	U(1.0, 2.0) %/°C

Fitted rather than sourced: k_G ~ U(1.5, 3) is calibrated on two simulated cases from one vendor, so it is an empirical correction, not a measured property. Derived from measurements but with an assumed shape: the DLC throttle Triangular(0.3, 0.6, 2.0) min and the generator start Triangular(0.15, 0.5, 3.2) min.

Limitations of the method

- **Validation is model against model.** The coverage harness judges the ensemble against perturbed copies of a model, and the thermal backtest is against *simulated* published curves. The only measured transients in the data are Tradat 2018 and Alkharabsheh 2017, and neither is run as a test yet. A measured site replay is the next real validation.
- **Coverage is short of target** when reality departs from the priors: 52 % against 75–85 %. The interval is labelled "model range" for that reason.
- **Lumped thermal model.** No hot spots, local recirculation or containment leakage; the well-mixed zone is the slow bound. The 4-minute WP179 restart is over-predicted by about 2.5–3.5 °C, and the loop volume is not identifiable.
- **Reliability data come from nuclear plants**, which are well maintained. The poorly maintained bound (generator fail-to-start 1.65e−2, about 7 times the INL mean) has not been run as a sensitivity case.
- **Water is not a fluid solver**: no momentum, no plenum flow, door thresholds only through zone elevations, and an unsourced drain rule.
- **The rig's vibration channel is qualitative.** It cannot verify mm/s or any ISO / AMCA zone, so vibration verdicts stay at the prior. The BV-3 thresholds recommended in the dossier are not yet in the fixture.
- **Nemotron's efficiency gain is unmeasured.** The estimator is unbiased whatever it proposes, but whether its regions beat the cross-entropy fallback at equal trial counts has not been tested.
- **Source caveats.** WP179 values were read by web-fetch and hand-transcribed; Table 2 was reconstructed from scrambled text and is not quoted. Opengate 2007 is known only from press coverage. ISO 20816-3 and ISO 14694 were not read directly.

Future work, from the dossier, not built

Ambient-temperature (DFW month × hour) and chiller-derate families for the heat-wave scenario; a four-mode leak-rate mixture for unobserved leaks and a drain-capacity prior with a blocked-drain probability; a two-node chilled-water loop for the 4-minute restart miss; Tradat and Alkharabsheh backtests as tests; BV-3 vibration thresholds in the fixture; probe settling on the rig and a 20-hold rig calibration to replace the evidence-budget assumptions; an efficiency comparison of plain Monte Carlo, cross-entropy and Nemotron proposals; out-of-prior coverage; fix-accuracy checks and browser timing for the ensemble; a Web Worker and a `predictFix` ensemble field.

SECTION 8

References

Retrieved 2026-11-07 unless noted (dossier §9). M = measured, S = simulated, V = vendor claim. Downloaded copies and their licences are listed in `research/SOURCES.md`.

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